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Centre No: Centre Name:

Code: Subject:

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2(d)

(i) [-0, 8] (only integers)

(ii)
```
public class RandomNumber;
import java.util.Scanner
public static int roll(int i){
    Scanner scan1 = new scanner
for (int i=1; i<=8; i++){
    scan1 = hextInt(8);
}
System.out.println("random:" + rnd.nextInt(8));
Scan1.close
}
```

1.

(a) Global variables are initialized before and and local variables are initialized in main. The advantage of global variable is that they are declared before main so they can be used everywhere but local variable will be destroyed after it was returned once.

(b) 9
2
9.4

(c)

a	a<20?	b
2	true	0
6	true	2
18	true	8
54	false	≥6

(d) There is no output at all. It is a runtime error, the value of int i can neve fit i<2

2.

(a)
=
=
=done
=
=done
It contains 5 lines.

Frankie — 23/40 (Q1 9/12 · Q2 6/15 · Q3 8/13)

Q2(d)(i) 0/2
nextInt(8) returns 0 to 7 - the argument is the upper bound and it is NOT included.

Q2(d)(ii) 0/2
roll() must take NO parameter and RETURN an int. Your version creates a Scanner, prints a value and never returns; nextInt(8) also gives 0-7. Correct body: return rnd.nextInt(8) + 1;

Q1(a) 2/3
Global vs local scope is roughly right (though locals are declared in any method, not only main), but the advantage you give is an advantage of GLOBAL variables. Advantages of local variables: the same name can be reused in different methods without clashing, and they are easier to debug.

Q1(d) 0/2
The body of a do...while always runs once BEFORE the condition is tested, so the program prints 3. There is no error.

Q2(a) 1/3
Follow-through (FT): '5 lines' is consistent with the output you wrote above, so the count earns 1 mark. The correct output is bar(2) -> '==' on one line, bar(4) -> '====', then 'done' on a third line: == / ==== / done - three lines.

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(b) public static int smallest(int a, int b, int c){
int[] arr = {a, b, c}
~~int a = 0~~
~~int~~ int MIN = a
~~int MAX = a~~
for (arr i = 0, i < arr.length; i++){
if (~~a~~ arr[i] > a){
~~int Max = arr~~ int MIN = arr[i];
}
}
System.out.println

(b) public static int smallest(int a, int b, int c){
int a = 0
int b = 0
int c = 0
System.out.println("a= ,b= ,c= ");
int[] arr = {a, b, c}
int MIN = a
for~~each~~ (arr i=0, i< arr.length; i++){
if (arr[i] < a){
int MIN = arr[i];
}
}
System.out.println("~~MIN=~~ smallest:" + MIN);

(c)
(i) 6
(ii) ~~Do not need to~~ Improve the reliability and reusability of the code, less likely to have error. Save the amount of code written.
2(d) is on the last page

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3.
(a) The amount of code written can be saved and improve the reusability of the code. Furthermore, it is easier for the teachers to calculate the average mark or give grades to students at a specific mark.

(b) 3
256

(c) public class marks;
int[] nums = {1, 3, 4}
~~int avg = 0~~ ⇒ double avg = 0;
int sum = 0;
int count = 0;
public static double average(int[] nums){
for(int i = 0; i < nums.length; i++){
~~if (i == nums[i])~~
count ++;
sum = sum + nums[i];
}
}
System.out.println("Avg:" + sum/count);

(d)
(i) ~~int[,] seats = {{3,1},{3,2}};~~
(ii) System.out.println(arr[2,3]);

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Q2(b) 1/4
The second version is the one marked (the first was crossed out). The outline is related, but: the method must RETURN the smallest - it prints; it redeclares a, b, c (compile error); and MIN is declared again inside the if, so it is never updated. Standard form: int min = a; if (b < min) min = b; if (c < min) min = c; return min;

Q3(b) 2/3
count = 3 is right; the total is 62 + 71 + 90 = 223, not 256.

Q3(c) 3/4
The loop and the division are right, but the method must RETURN the average (no println), and sum / count cannot be used: they are non-static fields read from a static method (compile error). Use a local double total and return total / nums.length.

Q3(d)(i) 0/2
Not answered - this line is crossed out. The declaration needs int[][] seats = new int[3][4] (rows first, then columns).

Q3(d)(ii) 0/1
The array is called seats, and row 2, column 3 is seats[1][2].